

- Q.28 Explain the need for periodical inspection of Airspeed indicator?
- Q.29 How does a RPM indicator work?
- Q.30 How is periodical inspection of pressure gauge done?
- Q.31 How does fuel flow gauge work?
- Q.32 Explain functioning of VSI with the help of a diagram.
- Q.33 Explain the functioning of manifold pressure gauge.
- Q.34 Describe various electrically operated instruments.
- Q.35 Classify the various instruments used in an aircraft.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is the need of an instrument? Describe the different types of instruments based on their working.
- Q.37 Explain the procedure for installation and replacement of any two of the Aircraft instruments.
- Q.38 Describe the different pressure and temperature gauge used in instruments.

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4th Sem / Aircraft Maintenance

Subject:- Aircraft Instruments

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the difference between the pitot pressure and the static pressure called?
- Atmospheric Pressure
 - Dynamic Pressure
 - Air Pressure
 - Cabin Pressure
- Q.2 The two additional radial markings on the airspeed indicator, one red and one blue, are associated with potential _____.
- Engine Failure
 - Landing Gear Failure
 - Radar Failure
 - Altimeter Failure
- Q.3 By correcting the indicated airspeed for nonstandard temperature and pressure, we obtain _____ airspeed.
- Actual
 - Virtual
 - True
 - Relative
- Q.4 The turn indicator indicates _____ motion of the airplane.
- Pitch
 - Yaw
 - Roll
 - Sideslip

- Q.5 Which of the following is the value of the atmospheric pressure, used to adjust the sub-scale of a pressure altimeter?
- a) Barometric setting b) Atmospheric setting
c) Altimeter setting d) Altitude setting
- Q.6 What is the frequency at which aircraft equipment's operate?
- a) 350 Hz b) 450 Hz
c) 400 Hz d) 40 Hz
- Q.7 What is the full form of ISA?
- a) International Stationary Atmosphere
b) Internal Standard Atmosphere
c) International Standard Atmosphere
d) Internal Stationary Atmosphere
- Q.8 What does the attitude indicator display?
- a) Aircraft's relation to the horizon
b) Aircraft's relation to the ground
c) Aircraft's relation to the wind flow
d) Aircraft's relation to the atmospheric pressure.
- Q.9 The altimeter is used to shows the aircraft's _____
- a) Altitude b) Airspeed
c) True speed d) Mach speed
- Q.10 Which of the following can replace a compass in aircraft?
- a) Heading indicator b) Turn indicator
c) Artificial horizon d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the application of R.P.M. indicator?
- Q.12 What is the importance of periodical inspection?
- Q.13 Why indicator maintenance is performed?
- Q.14 Where the leak test is done?
- Q.15 In an airplane where is the static port located?
- Q.16 What does turn and bank indicator show?
- Q.17 Name some electrically operated instruments.
- Q.18 What is the use of VSI?
- Q.19 What does attitude indicator show?
- Q.20 How is position error corrected?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the mechanical construction details of air speed indicator.
- Q.22 How is leak test done on an altimeter?
- Q.23 What are the chances of error in VSI?
- Q.24 What is the procedure for installation of aircraft instruments?
- Q.25 Describe a 5-hole pitot tube?
- Q.26 What are the different setting types of altitude measuring devices?
- Q.27 What is the replacement procedure of VSI?